

NEWCASTLE NOBBYS HEAD, Australia

WMO: 947740

Lat: 32.919S

Lon: 151.798E

Elev: 108

StdP: 14.64

Time Zone: 10.00 (AUS)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	45.8	47.5	31.4	25.8	58.5	33.6	28.4	57.5	43.8	60.8	36.4	58.4	12.4	300	0.562

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	7.3	87.8	67.2	82.6	67.2	79.0	68.8	74.3	78.3	73.2	77.0	72.2	76.0	19.4	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.2	123.8	75.9	72.0	118.9	75.1	71.0	114.7	74.4	37.9	78.4	36.9	77.4	36.0	76.0	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
35.4	31.1	27.6	DB	42.1	99.1	2.0	4.2	40.7	102.1	39.5	104.6	38.3	107.0	36.8	110.0	
			WB	38.9	76.3	1.6	1.7	37.7	77.5	36.7	78.5	35.8	79.5	34.6	80.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.1	72.5	72.6	71.2	67.0	61.9	57.9	56.4	57.8	61.8	65.1	67.7	70.3
	DBStd	6.87	3.72	3.32	3.31	3.53	3.57	3.29	3.16	3.99	4.61	4.88	4.54	3.94
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	1032	0	0	1	18	108	215	268	226	118	57	17	4
	CDD50	5528	698	632	658	510	369	236	198	243	355	467	532	630
	CDD65	1084	233	212	194	78	12	0	0	4	24	59	99	169
	CDH74	2929	675	538	369	128	16	0	1	20	140	287	343	412
	CDH80	737	192	84	58	20	0	0	0	2	30	99	126	126
(14) Wind	WSAvg	13.4	14.1	13.8	13.2	12.3	12.4	13.2	12.9	13.1	13.6	13.9	14.3	14.0
(15) Precipitation	PrecAvg	44.5	4.0	4.2	5.0	4.1	4.8	5.5	2.7	2.7	2.4	3.1	3.3	3.3
	PrecMax	70.9	13.1	22.0	13.3	13.3	17.4	16.1	10.6	8.8	7.7	7.3	8.0	8.9
	PrecMin	23.5	0.1	0.4	0.1	0.2	0.4	0.9	0.0	0.0	0.1	0.2	0.5	0.3
	PrecStd	10.3	2.9	3.7	3.2	3.5	3.4	3.4	2.1	2.2	1.7	1.9	2.2	2.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	96.4	89.9	86.7	82.9	76.2	70.6	71.4	77.5	84.4	88.9	91.5	92.3
		MCWB	69.6	72.2	70.1	65.9	62.3	60.6	57.5	60.1	61.6	63.2	65.8	67.8
	2%	DB	84.4	81.3	79.7	77.4	72.4	67.2	67.1	71.3	78.1	82.1	82.5	82.5
		MCWB	70.0	72.7	70.3	66.0	62.2	58.6	55.6	56.2	60.1	63.2	65.3	67.1
	5%	DB	78.5	78.3	77.0	74.5	70.1	65.2	64.6	67.7	73.2	75.7	76.5	77.5
		MCWB	70.9	72.0	70.5	66.2	62.3	57.9	54.9	55.6	58.4	62.5	66.2	69.0
	10%	DB	76.7	76.8	75.5	72.6	68.4	63.8	62.6	65.1	69.1	71.5	73.1	75.1
		MCWB	70.5	71.3	69.8	65.8	61.5	57.4	54.3	55.4	58.8	63.1	66.8	69.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	74.9	76.3	75.3	71.5	67.8	63.0	61.9	62.6	66.0	68.8	71.6	74.2
		MCDB	81.2	80.1	78.4	74.8	70.2	66.5	65.8	69.0	71.1	74.5	77.1	78.4
	2%	WB	73.6	74.6	73.3	69.5	66.0	61.7	59.4	60.7	64.3	67.3	70.1	72.1
		MCDB	77.9	78.1	76.5	73.2	69.1	64.8	62.9	66.0	69.3	72.0	74.9	76.1
	5%	WB	72.7	73.5	72.0	68.4	64.4	60.4	57.6	59.1	63.0	66.0	69.0	70.9
		MCDB	76.6	77.4	75.4	72.4	68.3	63.5	61.7	64.7	68.3	70.8	73.4	74.8
	10%	WB	71.8	72.3	70.9	67.2	63.0	59.2	56.3	57.5	61.6	64.8	67.8	69.8
		MCDB	75.7	75.8	74.6	71.5	67.2	62.4	60.7	63.1	67.0	70.0	71.9	73.7
(35) Mean Daily Temperature Range	5% DB	MDBR	7.8	7.3	8.3	10.5	12.0	11.1	12.8	13.6	13.0	11.2	9.4	8.4
		MCDBR	13.5	11.0	11.6	14.6	15.1	13.1	16.3	18.3	21.3	21.5	18.5	15.5
	5% WB	MCWBR	5.6	5.4	5.9	7.5	7.8	7.3	8.6	9.0	9.2	8.5	6.9	6.4
		MCWBR	10.1	9.2	9.8	11.6	11.5	10.7	12.1	14.0	14.8	12.8	11.9	10.9
(40) Clear-Sky Solar Irradiance	taub		0.395	0.386	0.355	0.333	0.316	0.310	0.300	0.309	0.344	0.363	0.386	0.383
		taud	2.455	2.504	2.582	2.616	2.616	2.625	2.638	2.601	2.497	2.472	2.450	2.477
	Ebn at Noon		299	296	294	284	273	267	277	288	293	299	300	304
		Edh at Noon	38	35	30	26	23	22	22	26	32	36	38	37
(44) All-Sky Solar Radiation	RadAvg		2094	1843	1551	1242	974	778	907	1201	1580	1823	1962	2111
	RadStd		134	152	134	98	75	65	75	73	84	146	180	164

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.56	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (29 neighbors)		+0.77	N/A	N/A	+1.39	N/A	N/A	-2	-153	+342	+185

Nomenclature: See separate page